Is There Any Point in Arguing?

A Logical Theory of Convergence, Split and the Inconsistent Core of Debate

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Abstract

Purpose. We give a formal instrument that answers, for a concrete dispute, four questions: will it converge to agreement; what is the minimal set of commitments that makes agreement impossible; is that set substantive or verbal; and how many rounds convergence takes. **Methods.** Opinions are modelled as deductively closed theories on a Tarski closure lattice, and the exchange of arguments as compatibility-constrained merging on an interaction digraph. The inconsistent core is identified with a minimal unsatisfiable subset (MUS); the verbal/substantive distinction is formalised through typed vocabulary alignments relative to an anchored (shared) vocabulary. The proofs use Craig interpolation, Reiter duality and SAT/MaxSAT machinery. **Results.** Consensus is reachable if and only if the profile is jointly — not merely pairwise — consistent; under B-connectivity it is reached within $(n-1)B$ rounds, and the bound is attained. From an arbitrary profile, under arbitrary graph sequences and arbitrary (including adversarial) tie-breaking, the dynamics halts within $n(n-1)$ changing rounds and oscillations are impossible. A dispute is substantive if and only if the conflict lies on the anchored fragment, which is decided by a single satisfiability test after maximal disambiguation; the verdict is provably symmetric in the parties. Unblocking a dispute is equivalent to hitting the family of MUS cores, and the MaxSAT mass of the conflict does not grow under admissible translations. A stopping rule follows from the value of information of the anchor test. **Conclusions.** Verbal/substantive diagnostics is strictly cheaper than full core enumeration — one satisfiability call against enumeration that is exponential in the worst case — and applies wherever an anchored vocabulary exists de facto: negotiation, strategic management, consulting, contract drafting.

Keywords: logic of disagreement; consequence operator; minimal unsatisfiable subset; verbal disputes; Craig interpolation; belief merging; opinion dynamics; value of information.

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1 Introduction

People spend a great deal of time and nerves on arguments that lead nowhere. Some of these are fruitless in substance: the parties are logically incompatible, and no exchange of reasons will reconcile them. Another part — apparently a substantial one — is fruitless only in appearance: the participants attach different meanings to the same words and are in fact talking about different things. The classical example of the second kind is the dispute *"a tree fell in a forest where nobody was: was there a sound?"*, which goes on until the parties realise that "sound" means for one of them vibrations of the air and for the other a heard experience.

The goal of this work is a formal instrument that answers, for a concrete dispute, four practical questions.

(Q1) **Will the dispute converge?** If reasons are exchanged, will the parties reach agreement?

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(Q2) **If not, what is the inconsistent core?** The minimal set of commitments because of which agreement is impossible.

(Q3) **Is that core substantive or verbal?** Real logical incompatibility, or a desynchronisation of meanings.

(Q4) **How long to wait if it does converge?** An estimate of the number of rounds.

What is synthesis and what is claimed as new. We aim at honest positioning. Items (Q1), (Q2) and (Q4) assemble mature results: convergence of averaging and accumulating dynamics [7, 12, 27], belief merging under a consistency constraint [18], and minimal unsatisfiable subsets as cores of inconsistency [20, 16]. Item (Q3) is the claim to novelty: to the best of our knowledge the distinction between verbal and substantive disagreements has not previously been operationalised algorithmically and dynamically, through the bundle of a MUS core with typed vocabulary translations relative to an anchor. The nearest predecessors — typed signature translations in belief-change theory [22], whose mechanism is the same but whose purpose is different (language independence of operators), and the model-theoretic distinction between disagreement and contradiction [29] — are discussed in Section 11, where we also separate the anchor criterion from forgetting and variable elimination. The philosophical criterion of a verbal dispute goes back to Chalmers [3]; we give it a computational form.

Contribution.

C1. A convergence criterion.
Consensus is possible if and only if the profile is *jointly* consistent; pairwise compatibility does not suffice (Example 3.2). Under B-connectivity the population converges to $T^* = \bigvee_k x_k(0)$ in at most $(n-1)B$ rounds, and the bound is attained (Theorem 3.3, Remark 3.9).

C2. Global stabilisation.
From any profile, under any graph sequence and any tie-breaking rule, the dynamics halts after at most $n(n-1)$ changing rounds; oscillations are impossible; the limit clusters admit a coalitional representation $\bigvee_{k\in C_i} x_k(0)$ (Theorem 3.8).

C3. The anchor criterion.
Substantivity of a dispute is equivalent to a conflict on the anchored fragment $\mathcal{L}_A$ and is decided by a single consistency test after maximal disambiguation; the proof goes through Craig interpolation and the invariance of the A-fragment, and the verdict is symmetric in the parties (Theorems 4.6, 4.9).

C4. Minimal concessions.
Unblocking a dispute is equivalent to a hitting set of the family of MUS cores (Reiter duality), and the MaxSAT mass of the conflict does not grow under admissible translations (Proposition 3.10, Lemma 4.12).

C5. The economics of a dispute.
An additive time estimate $\operatorname{diam}(G)+d_{\vdash}$ for bounded agents, and a rationality criterion for debate through the value of information of the anchor test (Proposition 5.1, Theorem 7.1).

Relation to earlier versions. This text is the result of several rounds of adversarial checking, and three corrections made in the course of it are of interest to the reader rather than merely to the author. First, a topological framing was abandoned: the consequence operator is not additive and is therefore not a topological closure (Remark 2.3); all the machinery below is lattice-and-closure machinery. Second, pairwise and joint compatibility are genuinely different

conditions, and for $n \geq 3$ only the joint one governs convergence (Example 3.2). Third, the correct bound on convergence time for bounded agents is additive rather than multiplicative (Remark 5.2).

2 Preliminaries

Definition 2.1 (consequence operator)**.** $\mathcal{L}$ is a countable set of formulas. $\mathrm{Cn}\colon 2^{\mathcal{L}} \to 2^{\mathcal{L}}$ satisfies Tarski's axioms: (E) $A \subseteq \mathrm{Cn}(A)$; (M) $A \subseteq B \Rightarrow \mathrm{Cn}(A) \subseteq \mathrm{Cn}(B)$; (I) $\mathrm{Cn}(\mathrm{Cn}(A)) = \mathrm{Cn}(A)$. In addition we assume *finitariness*,

$$\mathrm{Cn}(A) = \bigcup \{\mathrm{Cn}(A_0) : A_0 \subseteq A \text{ finite}\},$$

and the presence of a symbol $\bot$ with finitary inconsistency:

$$\mathrm{Cn}(A) = \mathcal{L} \iff \bot \in \mathrm{Cn}(A) \iff \exists\, \text{finite } A_0 \subseteq A \text{ with } \mathrm{Cn}(A_0) = \mathcal{L}.$$

Finally we assume *structurality* with respect to renamings: every sort-preserving morphism σ of the signature Σ (a substitution of symbols) transfers derivability, $\varphi \in \mathrm{Cn}(\Gamma) \Rightarrow \sigma(\varphi) \in \mathrm{Cn}(\sigma(\Gamma))$. All three properties hold in classical propositional logic and in many-sorted first-order logic.

A *theory* is a set T with $T = \mathrm{Cn}(T)$; their collection $\mathcal{T}$ is a complete lattice (a Moore family): $\bigwedge K = \bigcap K$ and $\bigvee K = \mathrm{Cn}\big(\bigcup K\big)$. A theory is *consistent* if $T \neq \mathcal{L}$; the set $\mathcal{T}_c := \mathcal{T} \setminus \{\mathcal{L}\}$ is closed under $\wedge$ but not under $\bigvee$.

Lemma 2.2 (minimisation of inconsistency)**.** *Every inconsistent set $\Gamma \subseteq \mathcal{L}$ contains a finite inclusion-minimal inconsistent subset.*

Proof. By finitary inconsistency Γ contains a finite inconsistent Γ_0. Among the finitely many subsets of Γ_0 choose an inconsistent subset of minimal cardinality; it is also inclusion-minimal, since a proper subset would have smaller cardinality. □

Remark 2.3 (why not topology). Cn is not additive ($p \wedge q \in \mathrm{Cn}(\{p, q\})$ but $p \wedge q \notin \mathrm{Cn}(\{p\}) \cup \mathrm{Cn}(\{q\})$) and $\mathrm{Cn}(\emptyset) \neq \emptyset$, so Cn is not a topological closure. Everything below is lattice-and-closure machinery; no topology is used or needed.

Definition 2.4 (agents, merging, mass of disagreement)**.** A finite population N, $|N| = n$, carries a measure μ; the state of an agent is $x_i(t) \in \mathcal{T}_c$; the interaction digraph is $G_t = (N, E_t)$, where an edge $(u, v) \in E_t$ means "v hears u in round t". Agents i, j are *compatible* if $\mathrm{Cn}(x_i \cup x_j) \neq \mathcal{L}$; a finite set of theories is *jointly consistent* if the union of all its elements is consistent. When updating, an agent merges its theory with some inclusion-maximal jointly consistent subgroup of what it has heard, containing itself (*compatibility-constrained merging*; the choice of subgroup is fixed by an arbitrary deterministic rule — a modelling choice, the logical analogue of bounded confidence [14]). For $\varphi \in \mathcal{L}$ the set of dissenters is $D_t(\varphi) = \{i : \varphi \notin x_i(t)\}$ and the mass is $m_t(\varphi) = \mu(D_t(\varphi))$.

3 Part I. Will the dispute converge?

Definition 3.1 (inconsistent core)**.** For a set of formulas Φ an *inconsistent core* is an inclusion-minimal $M \subseteq \Phi$ with $\mathrm{Cn}(M) = \mathcal{L}$; the collection of all cores is denoted $\mathrm{MUS}(\Phi)$, and for a pair of positions we write $\mathrm{MUS}(x_i, x_j) := \mathrm{MUS}(x_i \cup x_j)$. These are exactly the minimal unsatisfiable subsets. By Lemma 2.2 every core is finite, and $\mathrm{MUS}(\Phi) = \emptyset \iff \mathrm{Cn}(\Phi) \neq \mathcal{L}$. The dispute of a pair is *resolvable* if $\mathrm{MUS}(x_i, x_j) = \emptyset$.

Example 3.2 (pairwise compatibility is not enough)**.** Let $x_1 = \mathrm{Cn}(\{p \vee q\})$, $x_2 = \mathrm{Cn}(\{\neg p\})$, $x_3 = \mathrm{Cn}(\{\neg q\})$. All three pairs are compatible, but $\mathrm{Cn}(x_1 \cup x_2 \cup x_3) = \mathcal{L}$. The effect also arises in the Horn model of Section 8: under the background rule $p \wedge q \to r$ the positions $\{p\}$, $\{q\}$, $\{\neg r\}$ are pairwise conflict-free but jointly contradictory. Hence for a population with $n \geq 3$ the convergence condition is joint, not pairwise, consistency.

Theorem 3.3 (criterion of meaningfulness)**.** *Consider the dynamics of Definition 2.4.*

(i) ***Impossibility.*** *If* $\mathrm{MUS}(x_i(0), x_j(0)) \neq \emptyset$ *for at least one pair, then at no moment* t *is the population in consensus: agreement is unreachable without belief revision.*

(ii) ***Convergence.*** *If the initial profile is jointly consistent,* $T^* := \bigvee_k x_k(0) \neq \mathcal{L}$*, and the graph is* B*-connected (for every* t *the union* $\bigcup_{s=t}^{t+B-1} E_s$ *is strongly connected), then for all* i *the states* $x_i(t)$ *grow monotonically and*

$$x_i(t) = T^* \quad \text{for all } t \geq (n-1)B; \tag{1}$$

for a static strongly connected G *it suffices that* $t \geq \mathrm{diam}(G)$*.*

(iii) ***Structure of fixed points.*** *Let* X^* *be a fixed point and* U_i *the subgroup chosen by agent* i*. Then* $x_j \subseteq x_i$ *for all* $j \in U_i$*; in particular agents that choose each other have equal theories, and the population splits into clusters of mutually merged agents. On a static graph, an edge* (u, v) *between two distinct clusters is possible in a fixed point only if* x_u *is incompatible with the block chosen by* v*.*

Proof. (i) The dynamics is inflationary: an agent always belongs to its own merging group, so $x_i(t+1) \supseteq x_i(t)$ and, by induction, $x_i(t) \supseteq x_i(0)$ for all t. Suppose $\mathrm{Cn}(x_i(0) \cup x_j(0)) = \mathcal{L}$. If at moment t all states coincided with some $T \in \mathcal{T}_c$, then

$$T = \mathrm{Cn}(T) \supseteq \mathrm{Cn}\big(x_i(t) \cup x_j(t)\big) \supseteq \mathrm{Cn}\big(x_i(0) \cup x_j(0)\big) = \mathcal{L},$$

contradicting $T \neq \mathcal{L}$. The argument uses only inflationarity and is therefore valid for any inflationary dynamics, not only ours.

(ii) First the invariant $x_i(t) \subseteq T^*$. It holds at $t = 0$; if it holds at moment t, then the union of the theories of any subgroup heard lies in $T^* \neq \mathcal{L}$ and is therefore consistent, so the unique maximal jointly consistent subgroup consists of all those heard together with the agent itself, the merge coincides with the lattice join

$$x_i(t+1) = \bigvee_{j \in \mathrm{In}_i(t) \cup \{i\}} x_j(t) \subseteq T^*,$$

and the invariant is preserved. Next, flooding. Fix k and put $S_k(t) = \{i : x_i(t) \supseteq x_k(0)\}$. By inflationarity $S_k(t)$ is non-decreasing and $k \in S_k(0)$. Suppose $S_k(t) \neq N$. In the window $[t, t+B)$ the union graph is strongly connected, hence there is a path from $S_k(t)$ to the complement, and on it an arc $(u, v) \in E_s$, $s \in [t, t+B)$, with $u \in S_k(t) \subseteq S_k(s)$ and $v \notin S_k(t)$. In round s agent v merges everything it hears, so $x_v(s+1) \supseteq x_u(s) \supseteq x_k(0)$ and $v \in S_k(s+1)$. Thus in every window of length B the set S_k strictly grows until it equals N; after $(n-1)B$ rounds $S_k(t) = N$ for all k simultaneously, i.e. $x_i(t) \supseteq \bigcup_k x_k(0)$, whence $x_i(t) \supseteq T^*$ after closure. Together with the invariant this gives $x_i(t) = T^*$. For a static strongly connected G (with $B = 1$) the set S_k grows along BFS layers from k: after $\mathrm{ecc}(k)$ rounds $S_k = N$, and $\max_k \mathrm{ecc}(k) = \mathrm{diam}(G)$.

(iii) Being a fixed point means $x_i = \bigvee_{j \in U_i} x_j$, whence $x_j \subseteq x_i$ for all $j \in U_i$; under mutual choice, two inclusions and hence equality. Partitioning into classes by the symmetric-transitive closure of the mutual-choice relation gives clusters of equal theories. If $(u, v) \in E$ with u and v in distinct clusters, then $u \notin U_v$, and since U_v is an inclusion-maximal jointly consistent subgroup of what v hears, $U_v \cup \{u\}$ is not jointly consistent. □

Lemma 3.4 (merging as a CRDT)**.** *On a jointly consistent profile (i.e. when $\bigvee_i x_i \neq \mathcal{L}$) compatibility-constrained merging coincides with the lattice join $\bigvee = \mathrm{Cn}(\bigcup \cdot)$ and is therefore commutative, associative and idempotent, like the merge of a state-based CRDT [27]; the update is monotone and inflationary, and under B-connectivity all replicas converge to the least upper bound $T^* = \bigvee_i x_i(0)$ — the unique limit reachable from the given initial profile (strong eventual consistency).*

Proof. On a jointly consistent profile any subset of what is heard is consistent in union (monotonicity of Cn), so the maximal jointly consistent group is everything heard, and merging is the join; the semilattice properties are the properties of $\bigvee$ in any lattice. Inflationarity and monotonicity follow from (E) and (M). Convergence to T^* and its uniqueness from the given profile is Theorem 3.3(ii). Globally there are many fixed points — any constant profile is fixed; uniqueness refers to the limit of the trajectory from a given profile. □

Remark 3.5. The correspondence holds only on a jointly consistent profile: if some pair has MUS $\neq \emptyset$, naive union is absorbed by $\bot$, and the population behaves according to Theorem 3.3(i),(iii). Thus consistency of the union is exactly the condition of correct state-based CRDT merging, and the inconsistent core of two specifications is their minimal conflicting set of requirements. Example 3.2 is a warning: for $n \geq 3$ replicas one must check joint, not pairwise, compatibility.

The question of the existence of fixed points in the general regime, including inconsistent profiles, is answered positively and uniformly in graphs and choice rules. The key is the finiteness of the reachable state space.

Lemma 3.6 (coalitional representability)**.** *Along any trajectory of the dynamics of Definition 2.4, for all i and t there is a coalition $C_i(t) \subseteq N$ with $i \in C_i(t)$ and $x_i(t) = \bigvee_{k \in C_i(t)} x_k(0)$. Consequently all reachable states lie in the finite lattice $J := \{\bigvee_{k \in C} x_k(0) : \emptyset \neq C \subseteq N\}$, $|J| \leq 2^n - 1$.*

Proof. Induction on t. Base: $C_i(0) = \{i\}$. Step: if $U_i(t) \ni i$ is the merging group of agent i, then

$$x_i(t+1) = \bigvee_{j \in U_i(t)} x_j(t) = \bigvee_{j \in U_i(t)} \bigvee_{k \in C_j(t)} x_k(0) = \bigvee_{k \in C_i(t+1)} x_k(0), \qquad C_i(t+1) := \bigcup_{j \in U_i(t)} C_j(t),$$

by associativity and idempotence of the join; and $i \in C_i(t) \subseteq C_i(t+1)$. □

Lemma 3.7 (chain length in J)**.** *Every strictly increasing chain $T_0 \subsetneq T_1 \subsetneq \cdots \subsetneq T_m$ of elements of J has $m \leq n - 1$.*

Proof. Put $B_l := \{k \in N : x_k(0) \subseteq T_l\}$. Then $T_l = \bigvee_{k \in B_l} x_k(0)$: the inclusion $\supseteq$ holds because T_l is closed and contains every generator with index in B_l; the inclusion $\subseteq$ holds because the coalition representing T_l is contained in B_l. From $T_l \subsetneq T_{l+1}$ it follows that $B_l \subseteq B_{l+1}$, and strictly so: if $B_l = B_{l+1}$ we would have $T_l = T_{l+1}$. A strictly increasing chain of subsets of N starting with $|B_0| \geq 1$ has at most $n - 1$ steps. □

Theorem 3.8 (global stabilisation)**.** *Consider the dynamics of Definition 2.4 with an arbitrary initial profile (joint consistency is not required), an arbitrary sequence of graphs (E_t) and an arbitrary — deterministic or adversarial — rule for breaking ties when choosing a maximal jointly consistent group. Then:*

(a) the state of each agent strictly increases at most $n - 1$ times; there are at most $n(n-1)$ rounds in which at least one state changes; after the last change the profile is constant, so oscillations are impossible;

(b) on a static graph with a deterministic choice rule the profile reaches a fixed point of the update map in at most $n(n-1)$ rounds;

(c) the limit profile has the coalitional representation $x_i^\infty = \bigvee_{k \in C_i} x_k(0)$ with $i \in C_i$ and is invariant with respect to every graph occurring in (E_t) infinitely often.

Proof. (a) By inflationarity the trajectory of each agent is a non-decreasing sequence in J (Lemma 3.6), hence the number of its strict increases is bounded by the chain length in J (Lemma 3.7), i.e. by $n-1$. Every changing round contains at least one strict increase, and there are at most $n(n-1)$ increases in total; once they are exhausted, changes are impossible. (b) On a static graph with a deterministic rule the update is a map $F\colon J^n \to J^n$; a round without changes means $F(X) = X$, i.e. a fixed point, which is then preserved. Until the fixed point is reached every round is a changing one, and there are at most $n(n-1)$ of those. (c) The representation is Lemma 3.6 in the limit; by (a) the sequence stabilises. If a graph E occurs in (E_t) infinitely often, it also occurs after stabilisation, and a round with E does not change the profile — this is exactly invariance. □

Remark 3.9 (what is closed and what remains). Theorem 3.8 removes the questions of existence of fixed points and of oscillations: the dynamics always halts in a polynomial number of rounds, and the limit is computable by direct simulation ($n(n-1)$ rounds, each with n merges and consistency checks). What remains open is an a priori characterisation of the coalitions C_i — the number and composition of clusters as a function of the graph and of the MUS structure of the profile — without running the dynamics. The bound (1) is attained: on a cycle with n vertices $\operatorname{diam}(G) = \lfloor n/2 \rfloor$ and consensus is reached exactly at $t = \operatorname{diam}(G)$, which Section 8 confirms numerically ($n = 45$, $t = 22$); on a path with n vertices $\operatorname{diam}(G) = n-1$, so the bound $(n-1)B$ is attained already for $B = 1$.

Proposition 3.10 (how to unblock)**.** *Let the positions be given by finite bases: $x_i = \operatorname{Cn}(B_i)$, $x_j = \operatorname{Cn}(B_j)$, $\Phi := B_i \cup B_j$, $\operatorname{Cn}(\Phi) = \mathcal{L}$. For $H \subseteq \Phi$, giving up the premises H makes the dispute resolvable if and only if H is a hitting set of the family of cores:*

$$\operatorname{Cn}(\Phi \setminus H) \neq \mathcal{L} \iff H \cap M \neq \emptyset \text{ for every } M \in \operatorname{MUS}(\Phi).$$

The minimal concession is a minimum hitting set (MUS/MCS duality in the spirit of Reiter [25]); finding it is NP-hard already in the propositional case, and for a small number of cores it is solved by enumeration.

Proof. ($\Leftarrow$) Suppose H hits all cores but $\operatorname{Cn}(\Phi \setminus H) = \mathcal{L}$. By Lemma 2.2 the set $\Phi \setminus H$ contains a minimal inconsistent $M \subseteq \Phi \setminus H$; then $M \in \operatorname{MUS}(\Phi)$ and $H \cap M = \emptyset$ — a contradiction. ($\Rightarrow$) If $H \cap M = \emptyset$ for some $M \in \operatorname{MUS}(\Phi)$, then $M \subseteq \Phi \setminus H$ and $\operatorname{Cn}(\Phi \setminus H) \supseteq \operatorname{Cn}(M) = \mathcal{L}$. NP-hardness of the minimisation is the standard reduction from vertex cover to hitting set. □

Remark 3.11. The formulation is deliberately given for bases and not for closed theories: deleting a formula from a deductively closed set changes nothing, since the remainder derives it back. Belief revision in the closed model is therefore a contraction operation in the AGM sense, not set-theoretic subtraction.

4 Part II. Is the core verbal or substantive?

Here we drop the tacit assumption of Part I that everybody has a common language with common meanings. This part is the main claimed contribution.

Definition 4.1 (interpretations and anchored vocabulary)**.** The syntax is a many-sorted signature Σ with sorts S. Every agent carries its own interpretation: the same symbols may mean different things for different agents, while the naive union $x_i \cup x_j$ identifies symbols by spelling. We fix an *anchored vocabulary* $A \subseteq \Sigma$ — the logical constants plus those symbols whose referent the parties really regard as shared (in applications: the agreed terms); $\mathcal{L}_A$ is the set of formulas all of whose non-logical symbols lie in A. The set A is an input of the model, not derived from it.

Remark 4.2 (why an anchor is indispensable). Without an anchored vocabulary the criterion of verbality is trivial: one can rename the whole vocabulary of agent i into fresh symbols, the union becomes consistent, and then any dispute, including p against $\neg p$, is declared verbal. All the content rests on the requirement that a translation must not touch A.

Definition 4.3 (admissible alignment)**.** An *alignment* is a map τ of the vocabulary of agent i (total on the symbols occurring in x_i) which: (a) is the identity on A and does not send unanchored symbols into the anchor, $\tau|_A = \mathrm{id}$ and $\tau(\Sigma \setminus A) \cap A = \emptyset$; (b) preserves sorts; (c) is injective. Its action on a theory is symbol-wise translation of formulas followed by closure: $\tau[x] := \mathrm{Cn}(\tau(x))$, the image $\tau(x)$ itself need not be closed. The image may consist of fresh symbols (disambiguation) or fall into the vocabulary of agent j (identification of meanings). The set of admissible alignments is $\mathcal{A}(i, j)$.

Definition 4.4 (type of the core)**.** Let $\mathrm{MUS}(x_i, x_j) \neq \emptyset$. The dispute is *verbal* if $\exists\, \tau \in \mathcal{A}(i,j) : \mathrm{Cn}(\tau(x_i) \cup x_j) \neq \mathcal{L}$ — with the meanings outside A correctly matched, the conflict is removed — and *substantive* if $\forall\, \tau \in \mathcal{A}(i,j) : \mathrm{Cn}(\tau(x_i) \cup x_j) = \mathcal{L}$.

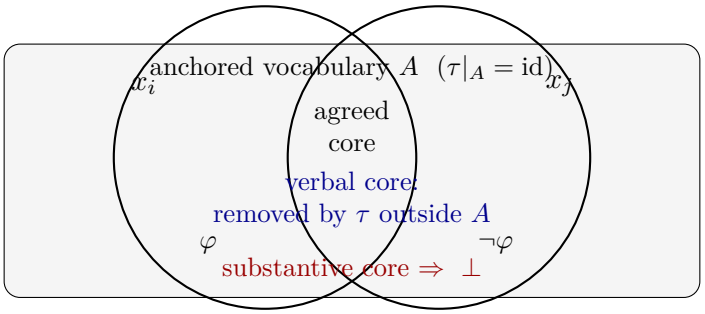


Figure 1: Anatomy of a dispute: the union of the positions stratifies into an agreed core, a verbal core (removed by a translation outside A) and a substantive core (on the anchor A, the conflict $\varphi, \neg\varphi$).

The key to the criterion is that an admissible translation does not change the A-fragment of a theory.

Lemma 4.5 (invariance of the A-fragment)**.** *For every theory x and every admissible τ,* $\mathrm{Cn}(\tau(x)) \cap \mathcal{L}_A = x \cap \mathcal{L}_A$.

Proof. ($\supseteq$) If $\varphi \in x \cap \mathcal{L}_A$ then $\tau(\varphi) = \varphi$, since all symbols of φ lie in A and $\tau|_A = \mathrm{id}$; hence $\varphi \in \tau(x) \subseteq \mathrm{Cn}(\tau(x))$. ($\subseteq$) Define a substitution σ: on $\mathrm{range}(\tau) \setminus A$ put $\sigma := \tau^{-1}$ (well defined by injectivity), and on the remaining symbols the identity; σ preserves sorts. Then $\sigma(\tau(\psi)) = \psi$ for all $\psi \in x$, i.e. $\sigma(\tau(x)) = x$. Let $\varphi \in \mathrm{Cn}(\tau(x)) \cap \mathcal{L}_A$. By structurality (Definition 2.1), $\sigma(\varphi) \in \mathrm{Cn}(\sigma(\tau(x))) = \mathrm{Cn}(x) = x$; but the symbols of φ lie in A, and by condition (a) of Definition 4.3 the set $\mathrm{range}(\tau) \setminus A$ is disjoint from A, so $\sigma(\varphi) = \varphi$ and $\varphi \in x \cap \mathcal{L}_A$. □

Theorem 4.6 (anchor criterion of substantivity)**.** *Let* Cn *be compact and have Craig interpolation with respect to shared non-logical symbols (true for classical propositional and for many-sorted first-order logic [6]), let x_i, x_j be consistent with* $\mathrm{MUS}(x_i, x_j) \neq \emptyset$*, and let $\tau^{\top}$ be the* maximal disambiguation*: the injective translation of all symbols of x_i outside A into pairwise distinct fresh symbols. Then the following are equivalent:*

(1) the dispute is substantive: every admissible τ leaves $\mathrm{Cn}(\tau(x_i) \cup x_j) = \mathcal{L}$*;*

(2) already $\tau^\top$ fails to restore consistency: $\mathrm{Cn}(\tau^\top(x_i) \cup x_j) = \mathcal{L}$*;*

(3) the conflict lies on the anchor: there is $\varphi \in \mathcal{L}_A$ with $\varphi \in x_i$ and $\neg\varphi \in x_j$.

In particular, verbality is equivalent to removability of the conflict by maximal disambiguation alone.

Proof. (1)⇒(2): $\tau^\top$ is admissible (identity on A, sort-preserving, injective, with a fresh image that does not touch A), so (2) is a special case of (1).

(2)⇒(3): the theories $\tau^\top(x_i)$ and x_j have shared non-logical symbols only in A, since the image of $\tau^\top$ is fresh. By compactness, from $\mathrm{Cn}(\tau^\top(x_i) \cup x_j) = \mathcal{L}$ we extract finite $\Gamma_i \subseteq \tau^\top(x_i)$, $\Gamma_j \subseteq x_j$ with $\neg \bigwedge \Gamma_j \in \mathrm{Cn}(\Gamma_i)$. By Craig interpolation there is χ all of whose non-logical symbols are shared by Γ_i and Γ_j, i.e. lie in A, with

$$\chi \in \mathrm{Cn}(\Gamma_i) \subseteq \mathrm{Cn}(\tau^\top(x_i)) \qquad \text{and} \qquad \neg\chi \in \mathrm{Cn}(\Gamma_j) \subseteq \mathrm{Cn}(x_j) = x_j.$$

By Lemma 4.5, $\chi \in \mathrm{Cn}(\tau^\top(x_i)) \cap \mathcal{L}_A = x_i \cap \mathcal{L}_A$. Put $\varphi := \chi$.

The degenerate case in which Γ_i and Γ_j share no non-logical symbol at all deserves a word, because there the interpolant is a logical constant. If $\chi \equiv \bot$ then $\bot \in \mathrm{Cn}(\tau^\top(x_i))$, so $\tau^\top(x_i)$ is inconsistent; but $\tau^\top$ is an injective sort-preserving renaming, and by structurality applied to $\tau^\top$ and to its inverse, $\mathrm{Cn}(\tau^\top(x_i)) = \mathcal{L}$ would force $\mathrm{Cn}(x_i) = \mathcal{L}$, contradicting the consistency of x_i. If $\chi \equiv \top$ then $\neg\top = \bot \in \mathrm{Cn}(\Gamma_j) \subseteq x_j$, contradicting the consistency of x_j. Hence χ is a genuine formula of $\mathcal{L}_A$ and the conclusion holds without exception.

(3)⇒(1): let $\tau \in \mathcal{A}(i,j)$ be arbitrary. By Lemma 4.5, $\varphi \in x_i \cap \mathcal{L}_A \subseteq \mathrm{Cn}(\tau(x_i))$, while $\neg\varphi \in x_j$; hence $\mathrm{Cn}(\tau(x_i) \cup x_j)$ contains φ and $\neg\varphi$ and by explosion equals $\mathcal{L}$. □

Corollary 4.7 (dominance of $\tau^\top$; a single SAT call)**.** *If any admissible alignment removes the conflict, then $\tau^\top$ removes it too. Therefore the verdict verbal/substantive is decided by a single consistency test on the set $\tau^\top(x_i) \cup x_j$; there is no need to enumerate $\mathcal{A}(i,j)$.*

Remark 4.8 (decidability, precisely)*.* Corollary 4.7 removes the enumeration of alignments but not the complexity of the logic itself. The diagnostic "is the dispute verbal?" is decidable only for a decidable Cn and finitely axiomatisable states — for instance in the propositional fragment, where it is a single NP satisfiability check, substantivity being its complement, coNP. In many-sorted first-order logic satisfiability is undecidable (Church), and on finite models (Trakhtenbrot) interpolation fails as well, so Theorem 4.6 does not apply there in form (2); the general case is undecidable. Sorts only shrink the space of translations, while the anti-triviality load is carried by A.

A natural question: is the model not too asymmetric, translating only side i? It is not.

Theorem 4.9 (symmetrisation of translation)**.** *Under the hypotheses of Theorem 4.6, let $\mathcal{A}(j,i)$ be the admissible alignments of the vocabulary of agent j (Definition 4.3 with the roles exchanged), and let $(\tau_i^\top, \tau_j^\top)$ be a pair of maximal disambiguations with pairwise distinct fresh images on both sides. The following are equivalent:*

(1) $\exists\, \tau \in \mathcal{A}(i,j) : \mathrm{Cn}(\tau(x_i) \cup x_j) \neq \mathcal{L}$ *(one-sided verbality);*

(2) $\exists\, (\tau_i, \tau_j) \in \mathcal{A}(i,j) \times \mathcal{A}(j,i) : \mathrm{Cn}(\tau_i(x_i) \cup \tau_j(x_j)) \neq \mathcal{L}$*;*

(3) $\mathrm{Cn}(\tau_i^\top(x_i) \cup \tau_j^\top(x_j)) \neq \mathcal{L}$*;*

(4) there is no $\varphi \in \mathcal{L}_A$ with $\varphi \in x_i$ and $\neg\varphi \in x_j$.

Proof. (1)⇒(2): the identity map is admissible, put $\tau_j := \mathrm{id}$. (2)⇒(4): contraposition. Let $\varphi \in x_i \cap \mathcal{L}_A$ and $\neg\varphi \in x_j$. For any pair (τ_i, τ_j), Lemma 4.5 applied to each side separately gives $\varphi \in \mathrm{Cn}(\tau_i(x_i))$ and $\neg\varphi \in \mathrm{Cn}(\tau_j(x_j))$, so the union explodes. (Disjointness of the images is not needed: only the inclusion $\supseteq$ of the invariance is used.) (4)⇒(3): contraposition. Let $\mathrm{Cn}(\tau_i^\top(x_i) \cup \tau_j^\top(x_j)) = \mathcal{L}$. The shared non-logical symbols of the two theories lie in A, both images being fresh and pairwise distinct, so, exactly as in step (2)⇒(3) of Theorem 4.6 — including the treatment of the degenerate interpolant — compactness and interpolation give $\chi \in \mathcal{L}_A$ with $\chi \in \mathrm{Cn}(\tau_i^\top(x_i)) \cap \mathcal{L}_A$ and $\neg\chi \in \mathrm{Cn}(\tau_j^\top(x_j)) \cap \mathcal{L}_A$; by Lemma 4.5, $\chi \in x_i$ and $\neg\chi \in x_j$, violating (4). (3)⇒(1): by Theorem 4.6, condition (1) is equivalent to (4), and (3)⇒(4) has just been proved by contraposition. The cycle is closed. □

Corollary 4.10 (invariance of the verdict)**.** *The verdict verbal/substantive does not depend on the direction of translation: whether side i translates its vocabulary, or side j, or both simultaneously, the diagnosis is the same and reduces to a single consistency test on the anchored fragments. In particular the procedure is symmetric in the parties and imposes on no one the role of the one who concedes the vocabulary.*

4.1 Working cases

Dispute	**Alignment τ outside A**	**Verdict**
Was there a sound (the tree in the forest)?	*sound* → {vibration of air; auditory experience}	verbal
Is a hot dog a sandwich?	*sandwich* → {bread + filling; two slices of bread}	verbal (borderline normative)
Freedom matters most	*freedom* → {negative; positive}	often verbal, a substantive residue is possible
Did the vaccine reduce mortality?	no admissible τ: the conflict is over A	substantive

Remark 4.11 (the mixed case, and the practical value). Usually a dispute is partly verbal: an optimal τ removes most of the conflict and a smaller substantive core remains. The value of the instrument is exactly this — to peel off the verbal layer and present the parties with the true, smaller crux: *you agree on most of it; what you really disagree about is only this.*

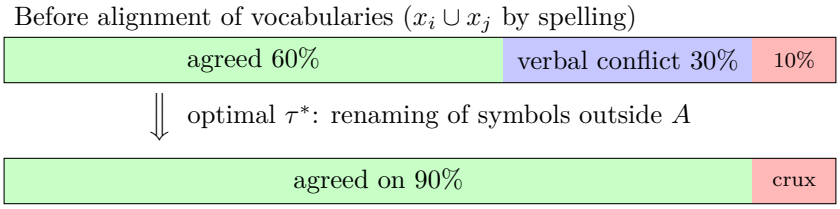


Figure 2: Peeling: the optimal translation τ^* moves the verbal mass into the agreed part, exposing a smaller substantive crux (the mixed case).

Lemma 4.12 (monotonicity of unblocking)**.** *Let τ be an admissible disambiguation, so that its image outside A consists of fresh symbols. Then:*

(a) renaming does not create new conflicts: if $\mathrm{Cn}(x_i \cup x_j) \neq \mathcal{L}$ then also $\mathrm{Cn}(\tau(x_i) \cup x_j) \neq \mathcal{L}$; a composition of disambiguations is a disambiguation;

(b) in the clausal representation the mass of the minimal concession does not grow: $\mu(\Phi^\tau) \leq \mu(\Phi)$, where $\Phi = \mathrm{cnf}(x_i) \cup \mathrm{cnf}(x_j)$, Φ^τ is the same after translation, and $\mu(\cdot)$ is the MaxSAT cost, i.e. the minimal number of clauses to be removed to regain consistency [2];

(c) *the residue that cannot be removed by any disambiguation cannot be removed by any admissible τ at all — this is the substantive core, and it is realised already by a single maximal disambiguation $\tau^\top$ (Corollary 4.7).*

Proof. (a) Contraposition: suppose $\mathrm{Cn}(\tau(x_i) \cup x_j) = \mathcal{L}$. The inverse substitution σ from the proof of Lemma 4.5 maps $\tau(x_i) \cup x_j$ to $x_i \cup x_j$ — on x_j it is the identity, since fresh symbols do not occur in x_j — and by structurality $\mathrm{Cn}(x_i \cup x_j) = \mathcal{L}$. A composition of injective, sort-preserving maps that are the identity on A and have fresh images has the same properties. (b) Let v be an assignment violating exactly $\mu(\Phi)$ clauses. Define v': on a fresh symbol $\tau(s)$ put $v'(\tau(s)) := v(s)$, and elsewhere $v' := v$. Every clause of Φ^τ is the image of a clause of Φ with the same truth value under (v', v), so v' violates at most $\mu(\Phi)$ clauses, whence $\mu(\Phi^\tau) \leq \mu(\Phi)$. (c) is a restatement of Theorem 4.6 and Corollary 4.7. □

Remark 4.13. An earlier version of this lemma asserted an inclusion of families $\mathrm{MUS}(\tau(x_i) \cup x_j) \subseteq \mathrm{MUS}(x_i \cup x_j)$. That statement is type-incorrect: the cores of the translated union consist of translated formulas and are not subsets of the original cores. The correct carriers of monotonicity are consistency (a) and the MaxSAT mass (b); it is exactly the mass that the procedure Peel of Algorithm 1 computes.

Remark 4.14 (duality with respect to the anchor). Enlarging the anchor, $A \subseteq A'$, shrinks the set of admissible alignments and therefore turns verbal cores into substantive ones — a special case of Theorem 4.6. Hence the resource for unblocking a dispute is not an extension of A but a controlled *de-anchoring* of a term: acknowledging it as ambiguous and licensing its translation.

5 Part III. How long to wait

Proposition 5.1 (convergence time)**.** *Let the dispute be resolvable and let the conditions of Theorem 3.3(ii) hold.*

(a) *For agents with full closure, the number of rounds until agreement is $R \leq \mathrm{diam}(G)$ on a static strongly connected graph, and $R \leq (n-1)B$ under B-connectivity.*

(b) *For* bounded *agents — those that in one round broadcast their current set of premises, merge what they hear, and perform one inference step — we have*

$$R \;\leq\; \mathrm{diam}(G) + d_{\vdash}, \tag{2}$$

where $d_{\vdash}$ is the logical depth, the maximal length of a derivation of the required consequences from the joint basis: the premises flood the graph in $\mathrm{diam}(G)$ rounds, after which each consequence is closed off locally in at most $d_{\vdash}$ steps.

Proof. (a) is proved in Theorem 3.3(ii). (b) The exchange of sets of formulas propagates every premise of the basis independently of derivation, so the flooding estimate from (a) applies to the premises verbatim; afterwards every agent has the whole basis, and a derivation of depth $\leq d_{\vdash}$ terminates in $d_{\vdash}$ rounds at one step per round. □

Remark 5.2. The multiplicative estimate $\mathrm{diam}(G) \cdot d_{\vdash}$ is not an upper bound in the edge case $d_{\vdash} = 1$: with the order "merge, then one inference step" the necessary budget is $\mathrm{diam}(G) + 1 > \mathrm{diam}(G) \cdot 1$. The additive form (2) is the correct one.

6 Part IV. Agreement is not truth

The fixed point of convergence is a consensus relative to the axioms and interpretations of the parties, not an objective truth. Truth is born in argument, but only for those who argue: the

limit T^* depends on the initial theories of the group, not on an external fact of the matter. This is a caveat against overclaiming, and a point of contact with the epistemology of disagreement [5].

It is worth locating our results against the Bayesian tradition, where the corresponding statements go the other way. Aumann's agreement theorem [1] shows that agents with a common prior and common knowledge of their posteriors cannot agree to disagree, and Geanakoplos and Polemarchakis [11] show that finitely many rounds of communication of posteriors force agreement in a finite partition model. In our setting there is no common prior and no external state space to condition on; the analogue of the common prior is the anchored vocabulary A together with joint consistency of the profile, and Theorem 3.3(i) says precisely that when this analogue fails — when some pair has a non-empty core — no amount of communication produces agreement. The Bayesian results are thus recovered in spirit, not as special cases: they concern conditioning on shared information, whereas Theorem 3.3 concerns the accumulation of commitments under a consistency constraint.

7 Part V. Is it worth arguing: rational stopping

Logic says whether a dispute will converge and in how many rounds; economics says whether it is worth paying for those rounds. Suppose agreement unblocks an action with utility $B > 0$, one round of exchange costs $c > 0$, and the number of rounds until agreement (if the dispute is resolvable) is bounded above by R from Proposition 5.1; we take the costs to be cR, a conservative estimate, since the real costs are no higher and lower thresholds are safe. The type of the core is latent: with probability π the core is verbal, in which case by Theorem 3.3 the dispute converges once the vocabularies are aligned, and with probability $1-\pi$ it is substantive.

Theorem 7.1 (rational stopping)**.** *The expected net value of blindly continuing the dispute equals* $\mathrm{NPV} = \pi B - cR$*. Then:*

(i) with a knowingly substantive core the resolving value of debate is zero while the costs are positive, so it is optimal to stop immediately; only diagnostic value remains;

(ii) arguing blindly is rational $\iff \pi B \geq cR$*, i.e.* $\pi \geq \pi^* := cR/B$*;*

(iii) the value of information of the anchor test (Theorem 4.6) equals

$$\mathrm{VoI} = \min\{\, cR(1-\pi),\ \pi(B - cR)\,\}, \tag{3}$$

and a test of cost c_d *is rational to perform before the dispute* $\iff c_d < \mathrm{VoI}$ *(the classical value-of-information scheme [15]).*

Proof. (i) With a conflict that is irremovable over A, agreement is unreachable (Theorem 4.6(1) and Theorem 3.3(i)): the benefit flow is 0, the costs are $cR > 0$. (ii) Linearity of NPV in π. (iii) Strategies without the test: argue blindly, with value $\pi B - cR$, or do not argue, with value 0; the better of the two gives $\max\{0, \pi B - cR\}$. The strategy "test, then act": we pay c_d; with a verbal verdict (probability π) we argue and get $B - cR$, with a substantive one we stop with 0; the total is $\pi(B - cR) - c_d$. The value of information is the difference ignoring c_d:

$$\mathrm{VoI} = \pi(B - cR) - \max\{0, \pi B - cR\} = \begin{cases} cR(1-\pi), & \pi B \geq cR, \\ \pi(B - cR), & \pi B < cR, \end{cases}$$

and both cases glue into (3), since $cR(1-\pi) \leq \pi(B - cR) \iff cR \leq \pi B$. The test is strictly profitable when $c_d < \mathrm{VoI}$. □

Remark 7.2. B is the instrumental value of agreement; expressive and status motives ("I want to be right") are zeroed out here. The threshold π^* links the structure of a dispute with its economics: a dense graph and shallow logical depth lower the entry bar, a sparse graph and deep derivations raise it. Using the upper bound R makes the thresholds conservative: if arguing pays off at cost cR, it also pays off at the actual, smaller cost.

8 Numerical illustrations

We model theories as sets of literals over Boolean variables with background Horn rules: the operator Cn is forward chaining, and compatibility is the absence of a variable assigned both ways. Because of Example 3.2 the simulator checks the joint consistency of the merged group, not only the pairwise one. All runs use fixed seeds and the code is given as supplementary material; the three experiments below are reported as run, and each is a check of a proved statement rather than an independent finding.

Graphs are rings with r random chords per vertex, so that the average degree is $k = 2 + 2r$; the population is $n = 45$ and the basis consists of 20 premises, each initially held by roughly 6% of the agents, with the seeding conditioned so that every premise has at least one source (otherwise the profile does not generate T^* and the mass of that premise trivially stays at 1).

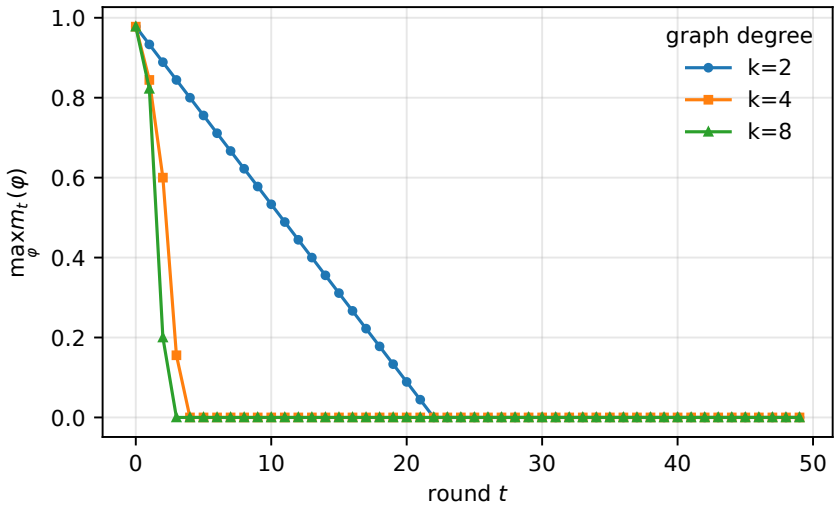


Figure 3: E1. The mass of disagreement vanishes in finite time, and the denser the graph the faster. On the ring ($k = 2$, diam $= 22$) consensus is reached exactly at $t = 22$, so the bound $R \leq \mathrm{diam}(G)$ of Proposition 5.1(a) is attained; for $k = 4$ and $k = 8$ the observed times are 4 and 3.

A separate experiment for Part II — measuring the share of verbal versus substantive cores on synthetic pairs of theories as a function of the anchored fraction $\alpha = |A|/n$ — is outside the scope of this paper and is stated as direction D10 in Section 13. What the anchor test costs relative to full core enumeration is settled analytically in Section 9: one satisfiability call against an output that is exponential in the worst case.

9 Implementation

All three questions of the paper reduce to calls of a SAT/MaxSAT solver over a CNF encoding of the theories: (Q1) will the dispute converge — a satisfiability check; (Q2) the inconsistent core — an unsat core plus minimisation to a MUS; (Q3) verbal or substantive — the anchor test as satisfiability after maximal disambiguation.

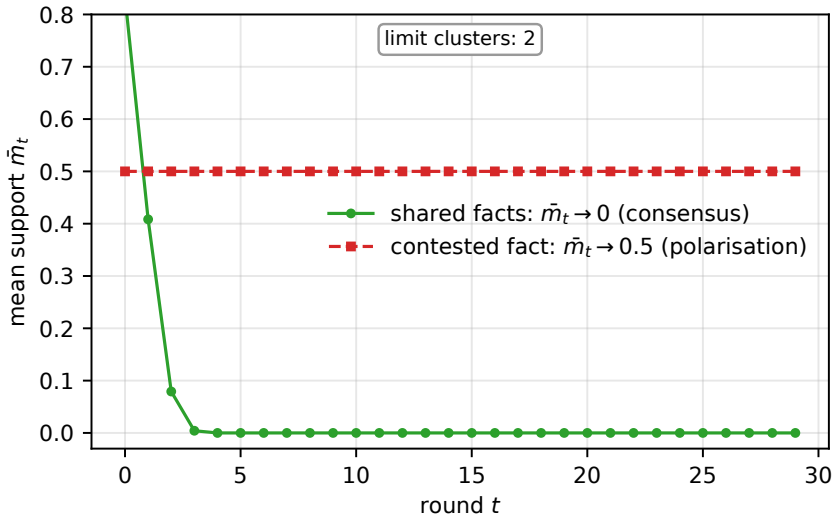


Figure 4: E2. Dichotomy: on shared facts the population reaches consensus, $\bar{m}_t \to 0$; on a contested fact (p against $\neg p$) it settles into stable polarisation, $\bar{m}_t \to 0.5$, with exactly two limit clusters — an instance of Theorem 3.3(i),(iii) and of the coalitional representation of Theorem 3.8(c).

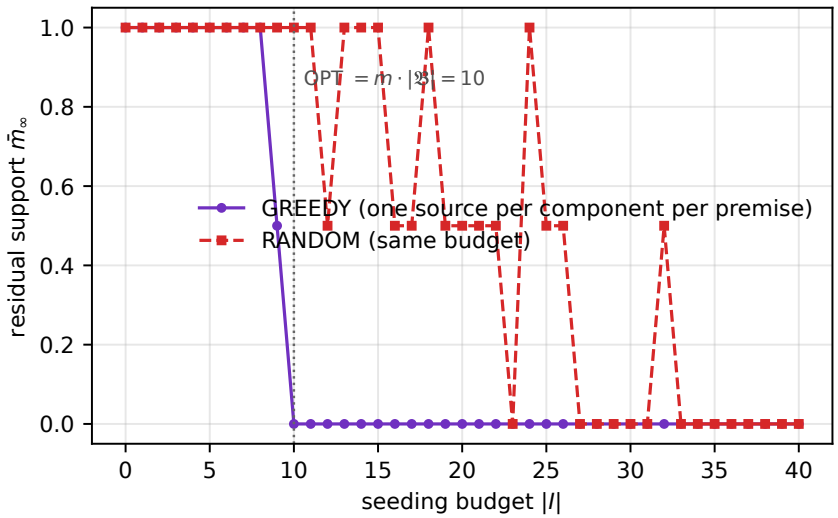


Figure 5: E3. Minimal seeding on a graph with $m = 2$ components and $|\mathfrak{B}| = 5$ premises: the greedy strategy — one source per component for each premise — attains the exact optimum $m \cdot |\mathfrak{B}| = 10$ and drives the residual support to zero at $|I| = 10$, whereas random seeding with the same budget wastes it and needs about four times as much. In this basic case the problem is polynomial; the NP-hardness of Proposition 3.10 concerns minimal concessions, not seeding.

9.1 Algorithm

We encode the theory of an agent as a set of clauses over a common signature Σ; the symbols $A \subseteq \Sigma$ are marked as anchored (frozen). Let $\Phi := \mathrm{cnf}(x_i) \cup \mathrm{cnf}(x_j)$. The key device of the anchor test is the maximal disambiguation $\tau^\top$: every unanchored symbol of the theory x_i is sent to a fresh symbol never seen before. The correctness of a single SAT call on $\tau^\top(x_i) \cup x_j$ is exactly Corollary 4.7: the surviving cores lie entirely over A and are removable by no admissible τ.

Algorithm 1 Diagnosing a dispute: compatibility, core, anchor test, residue

Require: theories x_i, x_j; anchored vocabulary A

function Compatible(x_i, x_j) ▷ (a) satisfiability
  $\Phi \leftarrow \mathrm{cnf}(x_i) \cup \mathrm{cnf}(x_j)$
  return Sat(Φ) ? Resolvable : Conflict ▷ Horn background: $O(|\Phi|)$
end function
function OneMUS(Φ) ▷ (b) deletion-based, MUSer style [20]
  $C \leftarrow$ UnsatCore(Φ) ▷ the solver's core: a superset of a MUS
  for all $c \in C$ **do**
    if $\neg$Sat($C \setminus \{c\}$) **then** $C \leftarrow C \setminus \{c\}$
    end if
  end for
  return C ▷ $O(|C|)$ SAT calls; C is one MUS
end function
function AllMUS(Φ) ▷ enumeration in MARCO style [21]
  enumerate MSS/MUS over the satisfiability map until exhaustion
  return MUS(Φ) ▷ the output is exponential in the worst case
end function
function AnchorTest(x_i, x_j, A) ▷ (c) the anchor test = one SAT call
  $\tau^\top \leftarrow \{\, s \mapsto \mathrm{fresh}(s) : s \in \mathrm{sym}(x_i) \setminus A \,\}$
  $\Phi^\top \leftarrow \mathrm{cnf}(\tau^\top(x_i)) \cup \mathrm{cnf}(x_j)$
  if Sat($\Phi^\top$) **then return** (Verbal, $\tau^\top$)
  else return Substantive ▷ surviving MUSes lie over A (Thm. 4.6)
  end if
end function
function Peel(x_i, x_j, A) ▷ (d) the mixed case — MaxSAT
  $\mu_{\mathrm{tot}} \leftarrow$ MaxSatCost(Φ) ▷ minimal concession without translation
  $\mu_{\mathrm{sub}} \leftarrow$ MaxSatCost($\Phi^\top$) ▷ irremovable residue; $\mu_{\mathrm{sub}} \le \mu_{\mathrm{tot}}$ by Lemma 4.12(b)
  return $(\mu_{\mathrm{sub}},\ \mu_{\mathrm{tot}} - \mu_{\mathrm{sub}})$ ▷ residue and the verbal mass removed by translation
end function

Complexity, precisely. Compatibility is NP in general and polynomial, $O(|\Phi|)$, on a Horn or 2-CNF background as in Section 8. Verbality, Sat($\Phi^\top$) by Corollary 4.7, is one NP call; substantivity is its complement, coNP. Computing one core, OneMUS, is in $\mathrm{FP}^{\mathrm{NP}}$ (a linear number of SAT calls); checking that C is a MUS is D^p-complete [23], and deciding whether a clause belongs to some MUS is Σ_2^p-complete [19]. As for all cores, the number of MUSes can reach $\binom{n}{\lfloor n/2 \rfloor}$; there is no polynomial-delay guarantee for enumeration, though MARCO [21] is efficient in practice and works in anytime mode. MaxSAT — the minimal concession, that is a minimum hitting set over all cores by Proposition 3.10 — is in $\mathrm{FP}^{\mathrm{NP}}$, the optimisation problem being NP-hard [2]. This is the precise sense of the claim that diagnosis is cheaper than full analysis: one NP call against an enumeration whose output alone is exponential.

9.2 Specification of the tool

Input	two theories x_i, x_j (typed literals/clauses: JSON `{sort,sym,neg}` or extended DIMACS with a symbol table) plus the vocabulary A (a list of frozen symbols `anchored:[...]`).
Output	a JSON report: `verdict` $\in$ {resolvable, verbal, substantive, mixed}; `cores`, the list of MUSes as sets of clause ids; `tau`, a partial τ (unanchored symbol $\mapsto$ fresh | aligned); `residual`, the substantive core over A with the masses μ_{sub} and $\mu_{\text{tot}} - \mu_{\text{sub}}$; `concession`, a minimum hitting set.
Export	optional: a DRAT certificate of inconsistency; GML of the core graph.
Backends	PySAT (Cadical/Glucose) for SAT; MUSer2/MARCO for cores; RC2/MaxHS for MaxSAT.

10 Applications

The model requires an anchored vocabulary A as input — and in negotiation, managerial and consulting practice this vocabulary exists de facto, which makes the theory non-vacuous exactly where it is useful.

Applied pipeline. In all the domains below the deployment is the same, and every step rests on a proved result: (s1) formalisation of the parties' positions into finite bases and fixing A — a glossary, the Definitions section, approved metrics; (s2) a consistency test of the union, deciding whether the dispute is resolvable (Theorem 3.3); (s3) if it is not, the anchor test by a single SAT call, verbal core against substantive (Theorems 4.6, 4.9); (s4) for a substantive residue, enumeration of MUSes and minimum hitting sets as a price list of concessions (Proposition 3.10); (s5) the decision argue / translate / concede / stop by the threshold π^* and VoI (Theorem 7.1). The output is not the facilitator's opinion but a reproducible verdict with a certificate: a concrete core, a concrete anchored formula φ, or a concrete set of concessions.

- **Negotiation and mediation.** A is what the parties have agreed not to dispute. The distinction between a verbal and a substantive core is a formal, diagnosable version of the principle "separate the people from the problem" [9, 24].
- **Strategic management.** A is the agreed metrics and KPIs. The diagnosis: a desynchronisation of definitions (verbal) or a genuine strategic trade-off (substantive).
- **Consulting.** The phase "let us agree on terms" is exactly the specification of A; the method then exhibits the true crux. A diagnostic stage formalised as a method.
- **Contracts and requirements.** A is the Definitions section; the tool flags where the conflict is carried by undefined terms.
- **An AI facilitator of meetings.** It listens to the dispute, extracts the core and reports: align the word X and you agree, or this is a real trade-off, here it is.

10.1 Law: completeness of definitions

Here A is the Definitions section. The coverage coefficient $\rho = |\Sigma_{\mathfrak{B}} \cap A|/|\Sigma_{\mathfrak{B}}| \in [0, 1]$ is the share of the operative terms $\Sigma_{\mathfrak{B}}$ that made it into the definitions. If $\rho = 1$ and the parties' positions are stated in the vocabulary $\Sigma_{\mathfrak{B}} \cup A$, then the contract generates no verbal cores, a specialisation of Theorem 4.6: every dispute is substantive, about a fact and not about words. The terms in $\Sigma_{\mathfrak{B}} \setminus A$ are points of latent verbal dispute, and the tool highlights them. Triage: a core that does not touch A (verbal) calls for a canon of interpretation, *contra proferentem* [26]; one that touches A (substantive) calls for an adversarial test of fact. Caveat: definiteness

of a term is not the same as generality of its referent (open texture), so $\rho = 1$ removes verbal cores only within the model.

10.2 Medicine: a threshold versus a recommendation

Two clinical guidelines share a measurable quantity $q \in A$ and a vocabulary of actions but define a diagnostic label H — say "hypertension" — by different thresholds $\theta_i \neq \theta_j$, with $H \notin A$. After eliminating the label, the conflict is verbal if and only if it is localised in the band between the thresholds, $q \in [\min(\theta_i, \theta_j), \max(\theta_i, \theta_j))$, where only the labels diverge; and substantive if on the common axis q there are contradictory actions over A, $do(a)$ against $do(\neg a)$. The harmoniser gives the physician a zone of pure definition and a residual clinical core — the subgroup of patients that requires a case conference or more data [29, 28].

10.3 Specifications and multi-agent LLMs

Two versions of a schema or of a set of requirements are theories S_i, S_j; by Lemma 3.4 jointly consistent ones merge as a CRDT, and for three or more replicas it is joint, not pairwise, consistency that must be checked (Example 3.2). The detector extracts $\mathrm{MUS}(S_i, S_j)$ — the minimal conflicting set of requirements — and by the anchor test relative to the agreed part of the schema decides between a verbal conflict (one name, different semantics, resolved by an alias) and a substantive one (calling for a revision of requirements). For a multi-agent LLM debugging dialogue this is a stopping condition: an empty core means let the agents converge; a non-empty one means print the core and the verdict instead of running one more round.

10.4 Mediation as an algorithm

The protocol, a formalisation of the "positions against interests" principle [9, 24]: (1) fix x_i, x_j and the anchor A; (2) compute the MUSes, and if there are none, collapse to agreement; (3) run the anchor test (Theorem 4.6); (4) for the verbal part, de-anchor the guilty term and split its meaning by a translation τ, the conflict mass not growing by Lemma 4.12; (5) iterate until a substantive residue; (6) on the residue, compute a minimum hitting set: you agree on most of it, and one point in each of these cores has to be conceded.

11 Relation to the literature

Opinion dynamics and belief alignment. Averaging models [7, 12] and bounded confidence [14]; belief merging with consistency postulates [18]; monotone convergence on a semilattice in the spirit of CRDTs [27]; network epistemology [30]; dynamic judgment aggregation [8]; the heterogeneous logical structure of opinions [10].

Inconsistency. MUSes as cores of inconsistency [20], their enumeration [21], the complexity of recognition [23, 19], inconsistency measures [16], MaxSAT [2] and the duality of conflicts and diagnoses [25].

Verbal disputes and meaning. The philosophical criterion [3]; the formal distinction between disagreement and contradiction through sets of models [29]; ambiguity of language and the blocking of consensus [13]; interpolation [6] as the technical core of the anchor criterion. Influence seeding: target set selection [4] and influence maximisation [17].

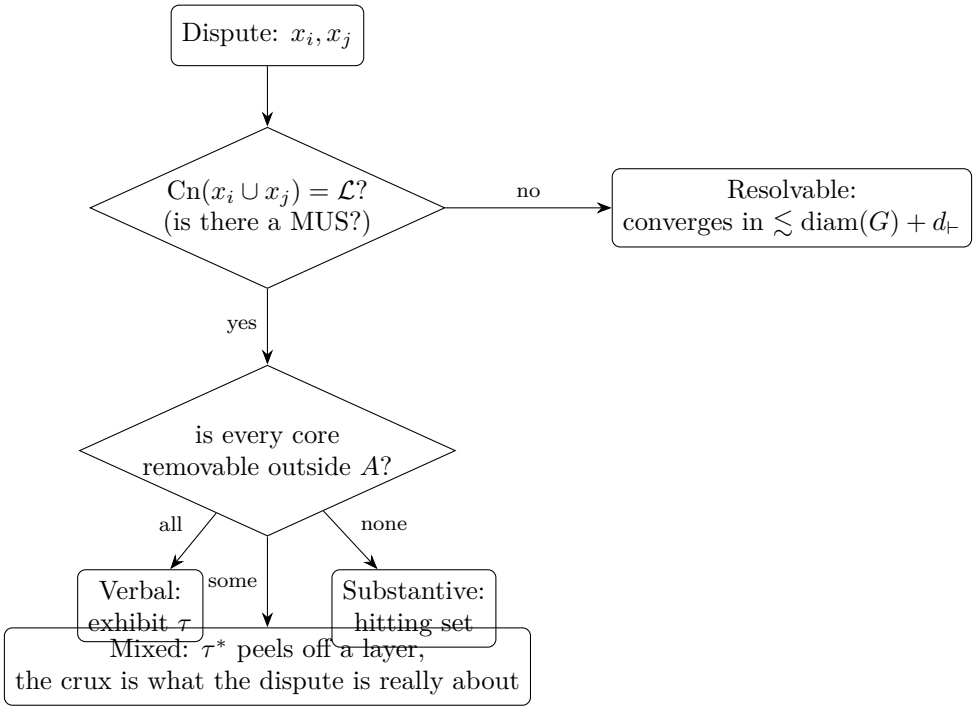


Figure 6: The decision diagram "is it worth arguing?": from a dispute to a verdict and a recommended action.

What the anchor criterion is not. The closest technical predecessor by mechanism is signature translation in propositional belief change [22], and the difference is worth stating explicitly, since it is the difference on which the novelty claim rests. There, translations serve to establish *language independence* of change operators: the object of study is an operator, and the desideratum is that renaming the signature should not alter its behaviour. Here translations are not a robustness requirement but a diagnostic device applied to a *pair* of theories, and the object of study is a property of that pair — the type of its inconsistent core — with the anchor A acting as the immune fragment that makes the property non-trivial.

The criterion should likewise not be confused with forgetting or variable elimination. Forgetting removes a symbol from a single theory and yields a weaker theory in a smaller signature; the anchor test performs an injective, sort-preserving *renaming* that preserves the full deductive content of x_i up to isomorphism of signatures — nothing is weakened, and by Lemma 4.5 the anchored fragment is preserved exactly. The two coincide nowhere except in the trivial case of a signature disjoint from A. That said, forgetting is the natural route to a constructive certificate of verbality in the propositional case, since an explicit A-fragment elimination replaces the interpolant by a computed formula; this is direction D5 below.

Our claim to novelty is not the individual ingredients but their bundle: MUS core × typed translation relative to an anchor × algorithmic diagnostics × opinion dynamics.

12 Limitations

- The anchored vocabulary A is a modelling assumption, an input, not derivable from the model; the content of Part II rests on it. Definiteness of a term does not guarantee generality of its referent (open texture).
- Theorems 4.6 and 4.9 are proved for compact logics with Craig interpolation (propositional, many-sorted first-order); on finite models interpolation fails and the status of the criterion is open.

- Decidability of the diagnostics holds only for the propositional or finitely axiomatisable fragment; in the first-order case the procedure is semi-decidable (Remark 4.8).
- Existence and reachability of the limit and the impossibility of oscillations are proved (Theorem 3.8); an a priori characterisation of the coalitions C_i — the number and composition of clusters as a function of the graph and of the MUS structure — without running the dynamics is open.
- The agents are idealised: full closure or one inference step per round; there is no noise, no lying, no strategic behaviour; B is the instrumental value of agreement without status motives.
- Countable populations and coalitional disputes (several against several) are outside the current model.
- The numerical section illustrates proved statements on synthetic Horn data; it is not an empirical validation on real debates. That is direction D10.

13 Further directions

A research programme, from nearer to farther; for each item the expected method is indicated.

D1. An a priori theory of clusters. Characterise the limit coalitions C_i of Theorem 3.8 combinatorially, through the hypergraph of MUS cores and the structure of G; conjecture: the clusters are the maximal jointly consistent communities with respect to reachability. Method: hypergraph colourings and hitting sets.

D2. Counting complexity. The number of clusters and fixed points: a suspicion of #P-completeness of counting and Σ_2^p-completeness of extremal questions, by analogy with [19].

D3. Graded beliefs. Replace theories by probabilistic or ranked states; merging becomes Bayesian combination with a consistency constraint; the question is an analogue of the anchor criterion, interpolation for probabilistic logic being known only fragmentarily.

D4. The paraconsistent regime. Remove explosion: merging in LP/FDE, where an inconsistent core is localised instead of infecting the theory; compare the clusters of the classical and the paraconsistent regimes.

D5. Finite models. Circumvent the failure of interpolation: uniform interpolation in the guarded fragment and EPR; or replace the interpolant by explicit A-fragment forgetting, which for the propositional case already yields a constructive certificate of verbality.

D6. Learning the anchor. Automatic extraction of A from a dialogue corpus with guarantees: robustness of the verdict to perturbations of A, the sensitivity $\partial\,\mathrm{verdict}/\partial A$ being a natural metric.

D7. Mechanism design for concessions. Minimum hitting sets as bargaining: sharing the cost of concessions (Shapley value or the core of a cooperative game on the MUS hypergraph), incentive compatibility of honest disclosure of bases.

D8. Strategic agents. Lying, concealment of premises, manipulation of the group-choice rule; the question is the robustness of Theorem 3.8, which survives adversarial tie-breaking but not adversarial states.

D9. Asynchrony and fault tolerance. Byzantine agents, node failures; the connection with self-stabilisation of distributed systems and CRDT practice.

D10. Empirics. Validation on corpora of real debates (structured disputes, court positions, code review), together with the synthetic phase-transition study of the share of substantive verdicts as a function of $\alpha = |A|/n$; measurable predictions: the share of verbal cores, the time to stabilisation, the accuracy of the forecast π^*.

D11. Multi-agent LLMs. Embed the anchor test into debate and verification protocols: detection of verbal divergence of the agents' ontologies before the budget is spent on arguing; a VoI-based planner of debate rounds.

14 Conclusion

We have given a logical model in which a dispute is an object with a decidable structure rather than a rhetorical process. Four results carry the argument. Convergence is governed by joint, not pairwise, consistency, and when it occurs it occurs within $(n-1)B$ rounds, a bound that is attained. Stabilisation is unconditional: whatever the graphs and whatever the tie-breaking, the dynamics halts within $n(n-1)$ changing rounds and never oscillates, so a dispute that has not converged has genuinely split rather than merely not yet finished. The type of the deadlock — verbal or substantive — is not a matter of interpretation but of a single satisfiability test relative to an anchored vocabulary, and the verdict does not depend on which side agrees to translate its terms. Finally, unblocking has a price list: the minimal concessions are the minimum hitting sets of the family of inconsistent cores, and whether the price is worth paying is decided by the threshold $\pi^* = cR/B$ and the value of information (3).

The practical consequence is narrow and, we think, real. Where the parties already share a vocabulary that they have agreed not to dispute — a glossary, a Definitions section, a set of approved metrics — a facilitator can replace an opinion about what the argument is "really" about with a certificate: this formula over the anchor, or this set of premises to be dropped. What the model does not deliver is equally clear: the anchor is an input rather than a discovery, the agents are idealised, and the fixed point is agreement, not truth. The most consequential next step is D6, learning the anchor from a corpus, since it is the anchor on which everything in Part II rests.

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Conflicts of interest. The author declares no conflict of interest.

Data availability. The code reproducing all simulations and figures of Section 8 is available from the author on request; all runs use fixed seeds and require no external data.

Declaration of generative AI in the writing process. During the preparation of this work the author used Claude (Anthropic) in order to check proofs, edit the text and translate it. After using this tool the author reviewed and edited the content as needed and takes full responsibility for the content of the publication. No AI system is listed as an author.

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